School Worksheet

Class 12 Mathematics, Chapter 2: Inverse Trigonometric Functions

Maximum marks: **43** **20** questions

Name _____ Date _____ Score _____ / 43

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

The principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$ is:

(A) $-\frac{\pi}{6}$ **(B)** $\frac{7\pi}{6}$ **(C)** $\frac{5\pi}{6}$ **(D)** $\frac{\pi}{6}$

2 MCQ — 1 mark

The value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ is:

(A) $\frac{7\pi}{6}$ **(B)** $\frac{5\pi}{6}$ **(C)** $\frac{\pi}{6}$ **(D)** $-\frac{\pi}{6}$

3 MCQ — 1 mark

The domain of the function $y = \sin^{-1}(2x)$ is:

(A) $[-1, 1]$ **(B)** $\left[-\frac{1}{2}, \frac{1}{2}\right]$ **(C)** $[-2, 2]$ **(D)** $[0, 1]$

4 MCQ — 1 mark

The range of the principal value branch of $\sec^{-1} x$ is:

(A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ **(B)** $(0, \pi)$

(C) $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ **(D)** $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

5 MCQ — 1 mark

The principal value of $\sec^{-1}\left(\sqrt{2}\right) + 2\csc^{-1}\left(-\sqrt{2}\right)$ is:

(A) $-\frac{\pi}{2}$ **(B)** $-\frac{\pi}{4}$ **(C)** $\frac{\pi}{4}$ **(D)** $\frac{\pi}{2}$

6 Assertion-Reason — 1 mark

Assertion (A): The domain of $y = \cos^{-1} x$ is $[-1, 1]$. **Reason (R):** The range of the principal value branch of $y = \cos^{-1} x$ is $[0, \pi] - \left\{\frac{\pi}{2}\right\}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

7 Short answer — 2 marks

Find the value of $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$.

Q No.	Question	Marks
Level 2		**16 marks**
8	Short answer Evaluate: $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right)+\tan^{-1}\left(\tan\frac{2\pi}{3}\right)$.	2 marks
9	Short answer Find the domain of $f(x)=\sin^{-1}\left(x^2-4\right)$.	2 marks
10	Short answer Find the domain of $f(x)=\sin^{-1}x+\cos^{-1}(x-1)$.	2 marks
11	Short answer Find the value of $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)+\tan^{-1}\left(\tan\frac{7\pi}{6}\right)$.	2 marks
12	Short answer Evaluate: $\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right]$.	2 marks
13	Short answer Find the value of $\tan^{-1}\left[2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right]$.	3 marks
14	Short answer Evaluate: $\sec^2\left(\tan^{-1}2\right)+\csc^2\left(\cot^{-1}3\right)$.	3 marks
Level 3		**19 marks**
15	Long answer Draw the graph of $y=\cos^{-1}x$ for its principal value branch. Using the graph, write its domain and range, and state whether the function is increasing or decreasing.	3 marks
16	Long answer Find the value of $\sin^{-1}\left(\sin\frac{4\pi}{5}\right)+\cos^{-1}\left(\cos\frac{4\pi}{3}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$.	3 marks
17	Long answer Find the domain and range of $f(x)=2\sin^{-1}(1-x)$.	3 marks
18	Long answer Write $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ in the simplest form.	3 marks
19	Long answer Write $\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}$, $0<x<\pi$, in the simplest form.	3 marks
20	Case study **Case Study:** A lighthouse is 20 m tall. A boat is at a horizontal distance of x m from the foot of the lighthouse. The angle of elevation θ of the top of the lighthouse from the boat is given by $\theta=\tan^{-1}\left(\frac{20}{x}\right)$. **(i)** Find θ when $x=20\sqrt{3}$ m. — 1 mark **(ii)** Write the domain and range of the principal value branch of $\tan^{-1}x$. — 1 mark **(iii)** Find the distance x of the boat when $\theta=\frac{\pi}{3}$. Also find the principal value of $\tan^{-1}\left(-\sqrt{3}\right)$. — 2 marks	4 marks

Answer key

Chapter 2: Inverse Trigonometric Functions, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $-\frac{\pi}{6}$
2	(B) $\frac{5\pi}{6}$
3	(B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$
4	(C) $[0, \pi] - \left\{\frac{\pi}{2}\right\}$
5	(B) $-\frac{\pi}{4}$
6	(C) A is true, but R is false
7	$\frac{2\pi}{5}$

Level 2

8	$-\frac{5\pi}{12}$
9	$\left[-\sqrt{5}, -\sqrt{3}\right] \cup \left[\sqrt{3}, \sqrt{5}\right]$
10	$[0, 1]$
11	$\frac{\pi}{3}$
12	1
13	$\frac{\pi}{4}$
14	15

Level 3

15	Domain $[-1, 1]$; range $[0, \pi]$; $y = \cos^{-1} x$ is a decreasing function.
16	$\frac{37\pi}{60}$
17	Domain $[0, 2]$; range $[-\pi, \pi]$
18	$\tan^{-1} x$
19	$\frac{x}{2}$
20	(i) $\theta = \frac{\pi}{6}$ (ii) Domain $\mathbb{R}$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (iii) $x = \frac{20\sqrt{3}}{3}$ m ≈ 11.55 m; $\tan^{-1}\left(-\sqrt{3}\right) = -\frac{\pi}{3}$